LAB PRACTICE № 1:
GETTING STARTED WITH JAVA

COMP1020: OOP and Data Structures **Week 01**

Lab Practice Submission Instructions:

- This is an individual lab practice and will typically be assigned in the computer laboratory.
- The lab computer is highly recommended over personal laptop.
- Your program should work correctly on all inputs. If there are any specifications about how the program should be written (or how the output should appear), those specifications should be followed.
- Your code and functions/modules should be appropriately commented. However, try to avoid making your code overly busy (e.g., include a comment on every line).
- Variables and functions should have meaningful names, and code should be organized into functions/methods where appropriate.
- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your code with other students to avoid plagiarism issues.
- Use the template provided to prepare your solutions (modified to provide your student ID, your name, etc).
- You should upload your .java file(s) to both the Canvas before the end of the laboratory session unless the instructor gave a specified deadline. No Canvas submission will get 0 point.
- Submit separate .java file for each lab problem with the exact file name specified in each lab question. For example: HelloWorld.java.
- How so submit .java file from eclipse: Right click on the Java file you want to export → Export → General → File System → Next → Select your Java file → Choose the destination folder → Finish
- Late submission of lab practice without an approved extension will incur the following penalties:

  1. No submission by the deadline will incur 0.25 point deduction for each problem (most of the problems are due at the end of the lab session).
  2. An additional 0.25 point per problem will be deducted for each day past the deadline.

3. The penalty will be deducted until the maximum possible score for the lab practice reaches zero (0%) unless otherwise specified by the instructor.

Part I: Setup Java Programming Environment

In this course, we are going to use "**Eclipse IDE**" software as the main IDE in the practical labs and exams. Following we will follow the instructions to install the Eclipse IDE for Java Developers:

1 - There are many ways but the prefer way is to download from this link: `https://www.eclipse.org/downloads/` ; it should direct you to your corresponding OS system (Windows/MacOS/Linux).

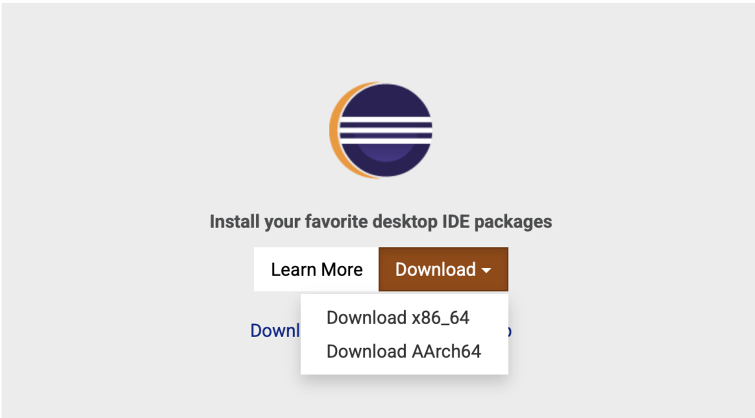

Figure 1: Get Eclipse IDE

2 - After download, run the file and select “**Eclipse IDE for Java Developers**” when asked (see Figure 2).

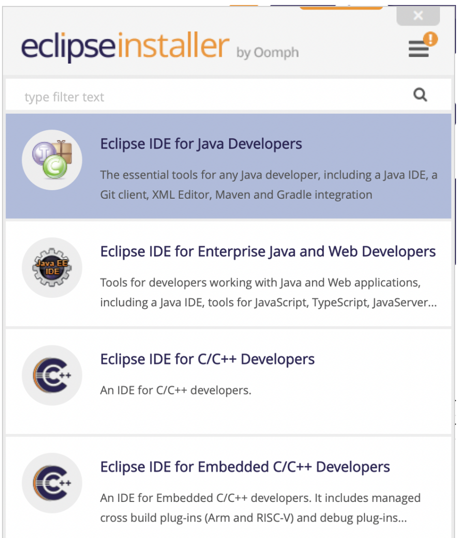

Figure 2: Select "Eclipse IDE for Java Developers".

3 - Next you can select where Java installation is located. Note that you should have internet

Figure 3: Choose the location and install the application.

connection during the installation.
Note: You are welcome to use other IDE as well (but you should also installed Eclipse IDE on your system just in case). Step by step to install Eclipse can be found the details below:

`https://www.eclipse.org/downloads/packages/installer`

Part 2: Problems

Problem 01: Sum of two integers

Write a program name `Lab01_P01.java`: Read two integers a and b from stdin, computes the sum of these integers and write the result to stdout.
Input

- Line 1: contains a $(1 \leq a \leq 1000000)$
- Line 2: contains b $(1 \leq b \leq 1000000)$

Output

- Write the sum $a + b$

Example See Figure 4

stdin	stdout
3 5	8

Figure 4: Example: Testcase Problem 01

stdin	stdout
Hello John, How are you? Bye,	HELLO JOHN, HOW ARE YOU? BYE,

Figure 5: Example: Testcase Problem 02

Problem 02: Convert a Text to upper case

Write a program name `Lab01_P02.java`: Given a TEXT, write a program that converts the TEXT to upper-case.

Input

- Line 1: contains the text

Output

- write the result to stdout

Example See Figure 5

Problem 03: Sum of elements of a sequence

Write a program and name it as `Lab01_P03.java`: Given a sequence of integers $a_1, a_2, \ldots, a_n$. Compute the sum Q of elements of this sequence.

Input

- Line 1: contains n ($1 \leq n \leq 1000000$)
- Line 2: contains $a_1, a_2, \ldots, a_n$ ($-10000 \leq a_i \leq 10000$)

Output

- Write the value Q

Example See Figure 6

Hint: You can quickly take a look at Java For Loop here:
`https://www.geekster.in/articles/java-for-loop/`

stdin	stdout
4 3 2 5 4	14

Figure 6: Example: Testcase Problem 03

stdin	stdout
1 1 8	NO SOLUTION

stdin	stdout
1 -7 10	2.00 5.00

stdin	stdout
1 -2 1	1.00

Figure 7: Example: Testcase Problem 04

Problem 04: Solve polynomial degree 2

Write a program and name it as `Lab01_P04.java`: Given 3 integers a, b, c, find all solutions to the equation $ax^2 + bx + c = 0$.

Input

- Line 1: contains a, b, c $(-1000 \leq a, b, c \leq 1000)$

Output

- Write NO SOLUTION if the given equation has no solution
- Write x_0 (2 digits after the decimal point) if the given equation has one solution x_0
- Write x_1 and x_2 with $x_1 < x_2$ (2 digits after the decimal point) if the given equation has two distinct solutions x_1, x_2

Example See Figure 7